

PET ROI Manager

Program Documentation

Prepared by Ryan Byrd

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1 Overview

The **PET ROI Manager** is a comprehensive Python application for semi-automated region-of-interest (ROI) analysis of PET phantom images. It provides an end-to-end workflow from DICOM series loading through SUV calculation, ROI placement, and results visualization.

This program was developed to streamline NEMA image quality testing and SUV verification on clinical PET/CT systems. It integrates Siemens-aware SUV computation logic, ensuring quantitative consistency even when vendor-specific decay corrections or scaling conventions differ.

2 Purpose

The goal of this program is to produce quantitative SUV and uniformity data from PET images in a reproducible, physics-driven manner. It allows the user to:

- Load a PET DICOM series directly from a scanner export.
- Select representative slices for uniformity and SUV analysis.
- Define phantom centers interactively using a crosshair cursor.
- Automatically generate the NEMA-specified ROI pattern around the phantom.
- Compute SUV metrics (mean, max, min) for each ROI.
- Export a CSV report and annotated overlay images.

3 Program Workflow

3.1 Step 1: Input and Output Selection

Upon execution, the program prompts the user to:

1. Select a folder containing PET DICOM images.
2. Choose a parent directory for saving results.
3. Enter a subfolder name for the output series.

All output files (CSV, PNG overlays, composite images, and optional RWVM sidecar) are stored in this newly created subdirectory.

3.2 Step 2: DICOM Loading and Rescaling

The DICOM series is read using `pydicom`. Each image is rescaled using the `RescaleSlope` and `RescaleIntercept` tags, except for Siemens images whose `Units` tag indicates BQML. In such cases, the voxel values are already in Bq/mL and no rescaling is applied.

Only valid 2D PET frames are included. The images are sorted by slice position (based on the `ImagePositionPatient` tag) and standardized to a consistent field of view (FOV = 250 mm).

3.3 Step 3: Slice Selection

The user is shown a grid of images and prompted to click four slices:

1. Three for uniformity measurement.
2. One for SUV and hot/cold sphere analysis.

The first three form the uniformity set; the fourth is used for the NEMA-style ring pattern ROI computation.

3.4 Step 4: Uniformity Measurement

After selecting the uniformity slices, the user clicks the phantom center. A circular ROI of diameter 180 mm is automatically placed around this point on all three slices.

For each slice:

- The mean SUV value is calculated within the ROI.
- Warnings are issued if the ROI extends beyond the image boundaries.

The average and per-slice uniformity means are later displayed on the composite output image.

3.5 Step 5: SUV Slice and ROI Geometry

The user then clicks three points on the SUV slice:

1. Phantom center.
2. The location of the 25 mm hot sphere.
3. The location of the 16 mm hot sphere.

These points define the phantom orientation. Using a predefined reference map (`REL_COORDS_MM`), the program automatically positions all seven standard NEMA ROIs:

25, 16, 12, 8, Bone, Air, and Water.

Each ROI is placed with correct rotational symmetry based on the user's clicks.

3.6 Step 6: SUV Calculation Logic

SUVs are calculated on a voxel-wise basis following these principles:

- Voxel intensities (Bq/mL) are rescaled using DICOM slope/intercept.
- The injected activity (`RadionuclideTotalDose`) is decay-corrected to the reference time (START or MID) unless `DecayCorrection = ADMIN`.
- The patient normalization mass (`PatientWeight`) is used to compute SUV as:

$$\text{SUV} = \frac{C_{\text{voxel}} \times M}{A_{\text{ref}}}$$

where C_{voxel} is the activity concentration, M is the normalization mass in grams, and A_{ref} is the decay-corrected injected activity.

- Siemens PET units are handled explicitly to avoid double decay-correction errors.

3.7 Step 7: Data Output

The following outputs are generated:

- **CSV file:** SUV mean, max, and min for each ROI.
- **Overlay PNG:** Displays all ROIs with numeric SUV labels.
- **Composite PNG:** 2×2 image grid showing the SUV slice and three uniformity slices.
- **Optional RWVM sidecar:** DICOM file encoding the voxel-to-SUV mapping for downstream quantitative tools.

Each image overlay is rendered in grayscale, inverted such that the phantom background appears white (for easier contrast). The overlay includes both ROI circles and text annotations.

4 Siemens-Aware SUV Handling

To ensure accurate quantification across vendors, the script includes logic specific to Siemens PET images:

- Siemens datasets with `Units = BQML` are interpreted as pre-scaled in Bq/mL.
- If `DecayCorrection = ADMIN`, activity normalization uses the injected activity A_0 directly (no decay correction).
- If `DecayCorrection = START` or `NONE`, the program decays A_0 forward to the acquisition start time.
- Optional handling of private tag (0071,1022) (Siemens Decay Correction DateTime) ensures accurate reference timing.

5 Output Example

- **CSV:** `SUV_results.csv`

```
ROI Label,SUVmean,SUVmax,SUVmin
25,5.8963,6.7321,5.2023
16,3.1745,3.5821,2.8992
...
```

- **Overlay:** `SUV_overlay.png` Displays circular ROIs with annotated SUV values.
- **Composite:** `SUV_composite.png` 2×2 grid layout showing the SUV map and three uniformity images.

6 Interpretation

- The uniformity slice means represent phantom background consistency (should be within $\pm 10\%$ across slices).
- The SUV values in the 25 and 16 mm spheres validate recovery coefficient performance.
- Cold region (Air, Bone, Water) mean and minimum values assess background suppression and spillover.

7 How to Use

1. Run `3. PET_ROI_Manager.py`.
2. Select the PET DICOM folder.
3. Choose the output folder and assign a name.
4. Select three uniformity slices and one SUV slice.
5. Click the center for uniformity, then center $\rightarrow$ 25 mm $\rightarrow$ 16 mm on the SUV slice.
6. Wait for results — the overlay and CSV files will open automatically.

8 Conclusion

The **PET ROI Manager** provides a fully guided, physics-compliant analysis pipeline for phantom-based PET SUV verification. By incorporating vendor-specific decay correction logic and reproducible ROI geometry, it standardizes SUV quantification across scanners and time-points—ensuring reliable baseline establishment and ongoing performance tracking.